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Mobile app login and device registration

Abstract

A mobile device for providing user authentication for a user accessing a website on a further device is provided. The mobile device includes a user interface; a display screen; a camera; a communication module; and a processor wherein the mobile device is configured to receive, via the user interface, an authentication request from a user, generate, using the processor, a machine-readable label generation code responsive to the authentication request, display, on the display screen, the generated machine-readable label generation code, capture, via the camera, an image of a machine-readable label, and verify, using the processor, that the machine-readable label in the captured image was generated using the machine-readable label generation code.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The application is the National Phase application of PCT Application No. PCT/EP2021/069493,

filed on Jul. 13, 2021 and titled “MOBILE APP LOGIN AND DEVICE REGISTRATION”. This application claims the benefit under 35 U.S.C. § 119 (a)-(d) of United Kingdom Patent Application No. 2011763.6, filed on Jul. 29, 2020 and titled “MOBILE APP LOGIN AND DEVICE REGISTRATION”. The above cited patent application is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present invention relates to a mobile device and a method of providing, using a mobile device, user authentication for a user accessing a website. The present invention also relates to a system comprising the above features, and a program configured to confer the above functionality.

BACKGROUND

(3) A variety of services may be accessible to a user accessing a website through a mobile device. It is common for websites to provide a level of security to prevent unauthorised users from accessing certain areas of a website, e.g. a secure dashboard. Prior art user authentication methods include the use of a registered username and password.

(4) However, prior art methods of providing user authentication can be inconvenient for a user. For example, it may be difficult for a user to accurately enter long usernames and passwords using a relatively small user interface (e.g. a touch screen of a mobile phone). Further, many websites recommend or require that passwords contain special characters (e.g. &, !, ?, etc.), capitalised letters, or numbers, which are often not displayed on a standard keyboard displayed on a mobile phone touch screen. The repeated switching between keyboards can be inconvenient to a user, and can increase the likelihood that a password is entered incorrectly. This can cause the user to fail the authentication process, and can potentially cause the user to be locked out of a website.

(5) The present invention attempts to address some of the drawbacks associated with prior art user authentication processes.

SUMMARY

(6) Aspects of the present invention are set out by the independent claims.

(7) According to a first aspect, there is provided a mobile device for providing user authentication for a user accessing a website on a further device, the mobile device comprising, a user interface, a display screen, a camera, a communication module, and a processor, wherein the mobile device is configured to receive, via the user interface, an authentication request from a user, generate, using the processor, a machine-readable label generation code responsive to the authentication request, display, on the display screen, the generated machine-readable label generation code, capture, via the camera, an image of a machine-readable label, and verify, using the processor, that the machine-readable label in the captured image was generated using the machine-readable label generation code. According to a second aspect, there is provided a system that includes the mobile device discussed above, a further device configured to access a website, and a physical or virtual server. According to a third aspect, there is provided method of using a mobile device having a user interface, display screen, a camera, a communication module and a processor whereby the method includes receiving, via the user interface, an authentication request from a user, generating, using the processor, a machine-readable label generation code responsive to the authentication request, displaying, on the display screen, the generated machine-readable label generation code, capturing, via the camera, an image of a machine-readable label, and verifying, using the processor, that the machine-readable label in the captured image was generated using the machine-readable label generation code. According to a fourth aspect, there is provided a computer-readable storage medium storing instructions that when executed by a processor of a mobile device having a user interface, a display screen, a camera, a communication module and the processor, configures the mobile device to perform a method comprising receiving, via the user interface, an authentication request from a user, generating, using the processor, a machine-readable label generation code responsive to the authentication request, displaying, on the display screen, the generated machine-readable label generation code, capturing, via the camera, an image of a machine-readable label,

and verifying, using the processor, that the machine-readable label in the captured image was generated using the machine-readable label generation code.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments will now be described, by way of example only, with reference to the accompanying drawings, in which:
- (2) FIG. 1 schematically illustrates an example of a system that includes a mobile device and further device connected to a server;
- (3) FIG. 2 schematically illustrates a further device suitable for use in the system of FIG. 1;
- (4) FIG. 3 schematically illustrates a server suitable for use in the system of FIG. 1; and
- (5) FIG. 4A and FIG. 4B illustrate an example of a flowchart describing steps of an authentication method.

DETAILED DESCRIPTION

- (6) Various embodiments, features, and aspects of the invention will be described in detail below with reference to the drawings. Each of the embodiments of the present invention described below can be implemented solely or as a combination of a plurality of the embodiments or features thereof where necessary or where the combination of elements or features from individual embodiments in a single embodiment is beneficial.
- (7) FIG. 1 illustrates a system **10**. The system comprises a mobile device **12**, a further device **14** and a server **16**. In this application, “server” includes a physical server, an application server, a cloud service provided on hosted or virtualised servers, or a true cloud service. The mobile device **12** is configured for two-way communication between the mobile device **12** and the server **16** as indicated by arrow A. The further device **14** is configured for two-way communication between the further device **14** and the server **16** as indicated by arrow B. In the example shown in FIG. 1, the two-way communication in each case takes place via wireless internet. However, any suitable communication protocol could be used in other examples.
- (8) The mobile device **12** comprises a display screen **18** and a user interface **20**. In the example shown in FIG. 1, the user interface **20** is separate to the display screen **18**. However, in other examples the user interface **20** could be integrated with the display screen **18**, e.g. as a touch screen. The mobile device **12** comprises a communication module **17**, via which the mobile device performs two-way communication with the server **16**, and a processor **19**. The mobile device **12** further comprises a camera **21** capable of capturing images that may be processed by the processor **19**. In the example shown in FIG. 1, the mobile device **12** is a mobile telephone. Other suitable mobile devices include tablet computers and other types of handheld computers.
- (9) The further device **14** comprises a display screen **22** and a user interface **24**. In the example shown in FIG. 1, the user interface **24** is separate to the display screen **22**. However, in other examples the user interface **24** could be integrated with the display screen **22**, e.g. as a touch screen.
- (10) In the example shown in FIG. 1, the further device **14** is a laptop computer. However, any suitable device (i.e. any device capable of accessing a website) could be used in practice. Examples include desktop computers, tablets and other mobile devices.
- (11) The server **16** comprises a communication module **26**, a processor **28** and a machine-readable label generation module. In the example shown in FIG. 1, the machine-readable label generation module is a QR-code generation module **30** capable of generating Quick Response (QR) codes. QR codes are well-known in the art as a type of machine-readable label containing non-volatile machine-readable content. However, in other examples the server **16** may comprise modules capable of generating any suitable machine-readable labels, such as linear barcodes, or images in

which machine-readable code may be embedded. Further, in the example shown in FIG. 1, the QR-code generation module 30 is shown as a module separate from the processor 28. However, in some examples, the QR-code generation module 30 could be integrated with the processor 28, e.g. as a processing section of the processor 28. The server 16 further comprises a memory 32 capable of hosting a website which is accessible by the further device 14 through two-way communication B.

(12) FIG. 2 illustrates a further device suitable for use as the further device 14 of FIG. 1. FIG. 2 shows selected standard components that are present in client computers such as the further device 14. The further device 14 is an example of an information processing apparatus, and in the embodiment of FIG. 1, it is a general-use laptop computer. The further device 14 has a CPU (Central Processing Unit) 201, a RAM (Random Access Memory) 202, a ROM (Read Only Memory) 203, a keyboard 204, a mouse 205, a display unit 206, an external storage device 207, and a network interface 208 which are all connected to each other via a bus.

(13) The CPU 201 is a standard processor such as those available from Intel® or AMD®. The RAM 202 is a conventional RAM and is used as a temporary storage area for instructions being processed by the CPU 201. The ROM 203 is a memory that stores certain applications such as a BIOS. The keyboard 204 and mouse 205 form input devices for the further device 14 in a conventional manner. The display unit 206 is a TFT display for providing output display for a user. The external storage device 207 is a removable USB hard disk drive. Network interface 208 is a set of standard components that allows the further device 14 to communicate with the server 16. Such devices are well known in the art and may include additional components (video cards etc.) or other components.

(14) FIG. 3 illustrates a server suitable for use as the server 16 of FIG. 1. FIG. 3 shows selected standard components that are present in servers. The server has a CPU 301, a RAM 302, a ROM 303, a storage 304, and a network interface 305 which are all connected each other via a bus. It is possible for those components of the server to be either distributed to multiple physical locations or integrated in a single housing. While the examples of FIGS. 1 and 3 describe physical servers, some or all of the services provided by such physical (e.g. local) servers may be provided by cloud services. Such services are typically accessed via the Internet. A cloud service can be considered a virtual server, due to the server functionality in fact being provided by any one of a plurality of online servers. Such a plurality of online servers is often referred to as a cloud, due to being located externally and remotely from the network. Thus, the mobile device 12 and the further device 14 may be configured to exchange information with a cloud service. A cloud service can also be considered as a software platform that provides an abstraction layer on top of the server (physical or virtual) layer. This is often referred to as true cloud (rather than hosted or virtualised servers). True cloud software is written to run on a software platform and to request or release resources as required.

(15) In the example of FIG. 3, the CPU 301 is a standard processor such as those available from Intel® or AMD®. The RAM 302 is a conventional RAM and is used as a temporary storage area for instructions being processed by the CPU 301. The ROM 303 is a memory that stores certain applications for use by the server 16, such as the BIOS. The storage 304 is a hard disk drive. Network interface 305 is a set of standard components that allows the server 16 to communicate with the mobile device 12 and further device 14.

(16) A method of providing user authentication using the system shown in FIG. 1 will now be described with reference to the flowcharts shown in FIGS. 4A and 4B. While the specific example of FIGS. 4A and 4B is described with respect to the use of a QR code to provide user authentication, it should be appreciated that other types of machine-readable label could be substituted (e.g. linear barcodes, images with encoded machine-readable data, etc.).

(17) When a user attempts to access a website hosted by the server 16 (or part of a website, e.g. a secure dashboard) through an app running on the mobile device 12, the server performs a check to

determine whether or not the user has logged in to the app. If it is determined that the user has not yet logged in to the app, a user authentication process is initiated. A method of providing user authentication begins with step **S101**.

(18) In step **S101**, a user inputs a login request via the user interface **20** of the mobile device **12**. The login request may be input to the app running on the mobile device **12** either manually, by the user entering a username and password, or by the user scanning a machine-readable label. These two options are presented to the user through the app.

(19) If the user selects the option of scanning a machine-readable label, the method moves on to step **S102**, in which the mobile device **12** generates a QR-generation code. This process may be carried out by the processor **19** of the mobile device **12**.

(20) Once the QR-generation code has been generated, the method moves on to step **S103**, in which the generated QR-generation code is displayed on the screen **18** of the mobile device **12**. In the present example, the QR-generation code is a six-digit number. However, in other examples, text string, alphanumeric codes, or any other suitable code may be generated. The QR-generation code may be displayed on the display screen **18** for a limited predetermined time period (e.g. one minute), in order to reduce the likelihood of an unauthorised user viewing the QR-generation code. The generated QR-generation code is temporarily stored in a memory of the mobile device **12**.

(21) After the QR-generation code has been displayed on the display screen **18**, the method moves on to step **S104**, in which the user, having viewed the QR-generation code on the display screen **18**, inputs the QR-generation code to the further device **14** via the user interface **24** of the further device **14**. In the example shown in FIG. **1**, the user interface **24** of the further device **14** is a keyboard. However, in other examples the user interface **24** could comprise a touch screen, a speech-to-text interface, or any suitable interface through which a user may input a code.

(22) To input the QR-generation code, a user first needs to access a code entry screen on the further device **14**. The code entry screen may be a section of the website that the user is attempting to access on the mobile device **12**. In order to access the code entry screen, a user needs to first input security credentials. The server **16** checks the input security credentials against a list of authenticated users stored in the memory **32** to confirm that the user is an authenticated user before displaying the code entry screen. The input of security credentials could be the provision of a username and password, the input of a PIN, the placement of the user's finger on a fingerprint scanner of the further device **14**, or any other suitable provision of security credentials.

(23) Once the user has input the QR-generation code to the further device **14**, the method moves on to step **S105**, in which the further device **14** transmits the QR-generation code entered by the user to the server **16** through two-way communication B. The server **16** receives the QR-generation code via the communication module **26**.

(24) Once the server **16** has received the QR-generation code, the method moves on to step **S106**, in which the server generates, in the QR-code generation module **30**, a QR-code using the QR-generation code. The QR-code generation module **30** encodes authentication information within the QR-code that may be later extracted from the QR-code by the processor of the mobile device **12**. The authentication information may indicate the user's identity based on the security credentials input to the further device **14** by the user in order to access the code entry screen. The QR-code generation module also encodes code-identification information within the QR-code that may later be extracted by the processor **19** of the mobile device **12**, said code-identification information indicating the specific QR-generation code that was used to generate the QR-code.

(25) Once the server **16** has generated the QR-code the method moves on to step **S107**, in which the server **16** transmits the QR-code to the further device **14** through two-way communication B.

(26) Once the further device **14** has received the QR-code from the server **16**, the method moves on to step **S108**, in which the further device **14** displays the received QR-code on the display screen **22**. The method then moves on to step C, where the method continues in FIG. **4B**.

(27) From step C, the method moves on to step **S109**, in which a user captures an image of the QR-

code being displayed on the display screen **22** of the further device **14** using the camera **21** of the mobile device **12**. The user may capture an image using an image capture screen displayed in the app running on the mobile device **12**. The app may display, on the display screen **18** of the mobile device **12**, a guide outline corresponding to the size and shape of a QR-code, such that when the app detects that the user has appropriately positioned the camera to fill the guide outline with the QR-code displayed on the display screen **22** of the further device **14**, the app performs an auto-focus process to ensure that the captured image of the QR-code is properly in focus. The app may be further configured to automatically capture an image when the app detects that the user has appropriately positioned the camera to fill the guide outline with the QR-code displayed on the display screen **22** of the further device **14**.

(28) Once an image of the QR-code has been captured by the user, the method moves on to step **S110**, in which the mobile device **12** performs a verification process on the captured image. The mobile device **12** performs the verification process by extracting code-identification information from the captured image of the QR-code, and comparing the extracted code-identification information to the QR-generation code stored in the memory of the mobile device **12**.

(29) The potential outcomes of the verification step are illustrated by step **S111**. A first outcome, illustrated by NO, occurs when the mobile device **12** determines that the QR-code in the captured image was not generated using the QR-generation code stored in the memory of the mobile device **12**. In this situation, the method moves on to step **S112**, in which the authentication process is immediately ended, e.g. by closing and/or locking the app running on the mobile device **12**. In some examples, the method may instead return to step **S109**, with the app prompting the user to re-try capturing the image of the QR-code displayed on the display screen **22** of the further device **14**. This process may be repeated a predetermined number of times (e.g. three attempts) before the method moves on to step **S112**.

(30) A second outcome, illustrated by YES, occurs when the mobile device determines that the QR-code in the captured image was generated using the QR-generation code stored in the memory of the mobile device **12**. In this situation, the method moves on to step **S113**, in which the mobile device **12** generates an authentication token. The authentication token contains authentication information extracted from the QR code.

(31) The mobile device **12** then transmits the authentication token to the server **16**. The server **16** compares the authentication information of the transmitted authentication token to the authentication information stored in the memory **32**, and confirms that they match one another.

(32) The authentication token may serve several purposes. Firstly, the authentication token may be used by the server **16** to authenticate the user, such that the user is permitted to access a secure section of the website hosted by the server **16** through the mobile device **12**. Additionally or alternatively, the authentication token may be used by the server **16** to register the mobile device **12** as an authenticated device, such that a secure section of the website hosted by the server **16** may be accessed through the app running on the mobile device **12** (e.g. regardless of the user of the mobile device).

(33) In some examples, the authentication may be time limited, such that the server **16** automatically revokes access to the secure section of the website through the mobile device **12** after the expiry of a predetermined time period.

(34) In some examples, multiple mobile devices could be registered as authenticated devices in a device registry stored in the memory **32** of the server **16**. For user convenience, devices stored in the device registry may be considered to be pre-authenticated by the server **16**, and so may be permitted to access the secure section of the website through an app running on the respective device without repeatedly undergoing the authentication process. However, devices may be added and/or removed from the device registry by a user by undergoing the authentication process.

(35) In some examples, the authentication process itself may be subject to time restrictions. For example, when the server **16** generates a QR code using the QR-generation code, a predetermined

period may be initiated (e.g. fifteen minutes) during which authentication tokens based on the generated QR code will be accepted by the server **16**. The server **16** may reject authentication tokens based on the generated QR code after the expiry of the predetermined time period, such that the user will need to re-start the authentication process.

(36) The above examples can also be realised by a computer of a system or apparatus (or devices such as a CPU or MPU) that reads out and executes a program recorded on a memory device to perform the functions of the above-described examples, and by a method, the steps of which are performed by a computer of a system or apparatus by, for example, reading out and executing a program recorded on a memory device to perform the functions of the above-described examples. For this purpose, the program is provided to the computer for example via a network or from a recording medium of various types serving as the memory device (e.g., a computer-readable medium such as a non-transitory computer-readable medium).

(37) While the present invention has been described with reference to embodiments, it is to be understood that the invention is not limited to the disclosed embodiments. The present invention can be implemented in various forms without departing from the principal features of the present invention. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. A system comprising: a mobile device for providing user authentication for a user accessing a website on a further device, the further device configured to access a website, and a physical or virtual server, the mobile device comprising: a user interface; a display screen; a camera; a communication module; and a processor; wherein the mobile device is configured to: receive, via the user interface, an authentication request from a user; generate, using the processor, a machine-readable label generation code responsive to the authentication request; and display, on the display screen, the machine-readable label generation code; wherein the further device is configured to: receive an input corresponding to the machine-readable label generation code, and transmit the machine-readable label generation code to the server; wherein the server comprises: a machine-readable label generation module; wherein the server is configured to: receive the machine-readable label generation code from the further device; generate, using the machine-readable label generation module, a machine-readable label, wherein the machine-readable label is encoded with information indicating the machine-readable label generation code; and transmit the machine-readable label to the further device; wherein the further device comprises a display screen, and wherein the further device is further configured to: receive the machine-readable label transmitted from the server, and display the machine-readable label on the display screen; and wherein the mobile device is further configured to: capture, via the camera, an image of the machine-readable label displayed on the display screen; and verify, using the processor, that the machine-readable label in the captured image was generated using the machine-readable label generation code.
2. A system according to claim 1, wherein the mobile device is further configured, upon verification that the machine-readable label in the captured image was generated using the machine-readable label generation code, to generate an authentication token using information extracted from the machine-readable label.
3. A system according to claim 2, wherein the mobile device is configured to transmit, via the communication module, the authentication token to the server.
4. The system according to claim 1, wherein the machine-readable label comprises a QR code.
5. A method of providing, using a mobile device that includes a user interface, a display screen, a camera, a communication module and a processor, user authentication for a user accessing a website on a further device, the method comprising the steps of: receiving, via the user interface, an authentication request from a user; generating, using the processor, a machine-readable label

generation code responsive to the authentication request; displaying, on the display screen, the machine-readable label generation code; receiving, at the further device, an input corresponding to the machine-readable label generation code, and transmitting the machine-readable label generation code to a physical or virtual server; receiving, at a machine-readable label generation module of the server, the machine-readable label generation code from the further device; generating, using the machine-readable label generation module, a machine-readable label, wherein the machine-readable label is encoded with information indicating the machine-readable label generation code; transmitting the machine-readable label to the further device; receiving the machine-readable label transmitted from the server; displaying the machine-readable label on a display screen of the further device; capturing, via the camera, an image of the machine-readable label; and verifying, using the processor, that the machine-readable label in the captured image was generated using the machine-readable label generation code.

6. The method according to claim 5, further comprising: upon verification that the machine-readable label in the captured image was generated using the machine-readable label generation code, generating, using the processor, an authentication token using information extracted from the machine-readable label.

7. The method according to claim 6, further comprising: transmitting, via the communication module, the authentication token to the server.

8. The method according to claim 5, wherein the machine-readable label comprises a QR code.

9. A non-transitory computer-readable storage medium storing instructions that, when executed by a processor of a mobile device, configure the mobile device, which includes a user interface, a display screen, a camera, a communication module and the processor, to execute a method to provide user authentication for a user accessing a website on a further device, the method comprising: receiving, via the user interface, an authentication request from a user; generating, using the processor, a machine-readable label generation code responsive to the authentication request; displaying, on the display screen, the machine-readable label generation code; receiving, at the further device, an input corresponding to the machine-readable label generation code, and transmitting the machine-readable label generation code to a physical or virtual server; receiving, at a machine-readable label generation module of the server, the machine-readable label generation code from the further device; generating, using the machine-readable label generation module, a machine-readable label, wherein the machine-readable label is encoded with information indicating the machine-readable label generation code; transmitting the machine-readable label to the further device; receiving the machine-readable label transmitted from the server; displaying the machine-readable label on a display screen of the further device; capturing, via the camera, an image of the machine-readable label; and verifying, using the processor, that the machine-readable label in the captured image was generated using the machine-readable label generation code.
